DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
January 2013
ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number** (**not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. (a) Show that the group of automorphisms of $\mathbb{Z}/221\mathbb{Z}$ has no element of order 5. (*Note:* $221 = 13 \times 17$.)
(b) Show that a group of order $1105 = 5 \times 13 \times 17$ must be cyclic.

2. Let $A \in M_3(\mathbb{C})$ be a 3×3 matrix with minimal polynomial X^3. Let $B \in M_4(\mathbb{C})$ be a 4×4 matrix with minimal polynomial X^4. The matrix $A \otimes B$ acts on $\mathbb{C}^3 \otimes \mathbb{C}^4 \simeq \mathbb{C}^{12}$ by $(A \otimes B)(v \otimes w) = Av \otimes Bw$.
(a) Find the minimal polynomial of $A \otimes B$ and justify your answer.
(b) Find the dimension of the kernel of $A \otimes B$.

3. Let R be a principal ideal domain and let a, b be nonzero elements of R that are not units. The set of R-homomorphisms

$$H = \mathrm{Hom}_R\,(R \oplus R/aR,\ R \oplus R/bR)$$

from $R \oplus R/aR$ to $R \oplus R/bR$ is an R-module, where the action of R is given by $(rf)(x) = r(f(x))$ (with $r \in R$ and $f \in H$) and the addition is pointwise addition of functions. Describe the decomposition of H as a direct sum of cyclic R-modules. Justify your answer.

4. Let K be a field.
(a) Let L/K be an algebraic extension. Let R be a ring with $K \subseteq R \subseteq L$. Show that R is a field.
(b) Suppose A is a commutative ring containing K and $\phi : A \to K[X]$ is a ring homomorphism that is the identity on K. Let M be a maximal ideal of $K[X]$. Show that $\phi^{-1}(M)$ is a maximal ideal of A.

5. Let K be a field of characteristic 0 such that every odd degree polynomial $f(X) \in K[X]$ has a root in K.
(a) Let L/K be a finite Galois extension. Show that $[L : K]$ is a power of 2.
(b) Let L/K be a finite extension. Show that $[L : K]$ is a power of 2.

6. Let $G \simeq (\mathbb{Z}/4\mathbb{Z}) \oplus (\mathbb{Z}/2\mathbb{Z})$ be the group generated by a and b with relations $a^4 = b^2 = 1$ and $ab = ba$. Let

$$A = \begin{pmatrix} i & 0 \\ 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 0 \\ 1+i & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(a) Show that there are matrices U and V such that $UAU^{-1} = S$ and $VBV^{-1} = T$.
(b) Show that the map $\rho : G \to GL_2(\mathbb{C})$ satisfying $\rho(a^j b^k) = A^j B^k$ is a well-defined representation of G.
(c) The homomorphism $\pi : G \to GL_2(\mathbb{C})$ satisfying $\pi(a^j b^k) = S^j T^k$ is a well-defined representation of G (do not show this; the proof is similar to (b)). Show that ρ and π are not equivalent representations of G.